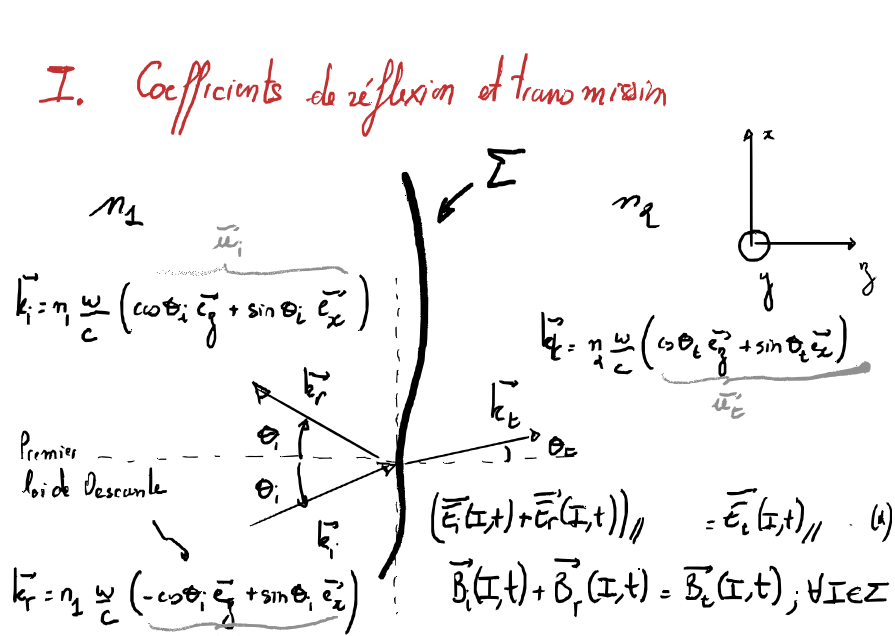
Explication TP He-Ne

I. Coefficients de réflexion et transmission



$\vec{k_i} = n_1 \frac{\omega}{c}\left(\cos\theta_i \vec{e_z} + \sin\theta_i \vec{e_x}\right)$

$\vec{k_t} = n_2 \frac{\omega}{c}\left(\cos\theta_t \vec{e_z} + \sin\theta_t \vec{e_x}\right)$

$\vec{k_r} = n_1 \frac{\omega}{c}\left(-\cos\theta_i \vec{e_z} + \sin\theta_i \vec{e_x}\right)$

$\left(\vec{E_i}(I,t)+\vec{E_r}(I,t)\right)_{//} = \vec{E_t}(I,t)_{//}$ (1)

$\vec{B_i}(I,t)+\vec{B_r}(I,t) = \vec{B_t}(I,t)\ ;\ \forall I \in \Sigma$

or pour des ondes planes $\vec{B} = \frac{\vec{k}\wedge\vec{E}}{\omega} = \frac{n}{c}\vec{u}\wedge\vec{E}$

donc : $n_1 \vec{u_i}\wedge\vec{E_i} + n_1 \vec{u_r}\wedge\vec{E_r} = n_2 \vec{u_t}\wedge\vec{E_t}$

$$n_1\left(\cos\theta_i \vec{e_z}\wedge\left(\vec{E_i}(I,t)-\vec{E_r}(I,t)\right) + \sin\theta_i \vec{e_x}\wedge\left(\vec{E_i}(I,t)+\vec{E_r}(I,t)\right)\right) = n_2\left(\cos\theta_t \vec{e_z}\wedge\vec{E_t}(I,t) + \sin\theta_t \vec{e_x}\wedge\vec{E_t}(I,t)\right)$$

$$\rightarrow \left.\begin{array}{l}\vec{e_z}\wedge\left[n_1\cos\theta_i\left(\vec{E_i}(I,t)-\vec{E_r}(I,t)\right) - n_2\cos\theta_t\vec{E_t}(I,t)\right] \\ + \vec{e_x}\wedge\left[n_1\sin\theta_i\left(\vec{E_i}(I,t)+\vec{E_r}(I,t)\right) - n_2\sin\theta_t\vec{E_t}(I,t)\right] = 0\end{array}\right\}(**)$$

On regarde dans le plan ⊥ à $\vec{e_z}$ (tangencielle) on fait $\vec{e_z}\wedge$ à (**) et $\vec{a}\wedge(\vec{b}\wedge\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$

$$\rightsquigarrow \left(\left[n_1\cos\theta_i\left(\vec{E_i}(I,t)-\vec{E_r}(I,t)\right) - n_2\cos\theta_t\vec{E_t}(I,t)\right]\cdot\vec{e_z}\right)\vec{e_z} - \left[n_1\cos\theta_i\left(\vec{E_i}(I,t)-\vec{E_r}(I,t)\right) - n_2\cos\theta_t\vec{E_t}(I,t)\right] + \left(\left[n_1\sin\theta_i\left(\vec{E_i}(I,t)+\vec{E_r}(I,t)\right) - n_2\sin\theta_t\vec{E_t}(I,t)\right]\cdot\vec{e_z}\right)\vec{e_x} = 0$$

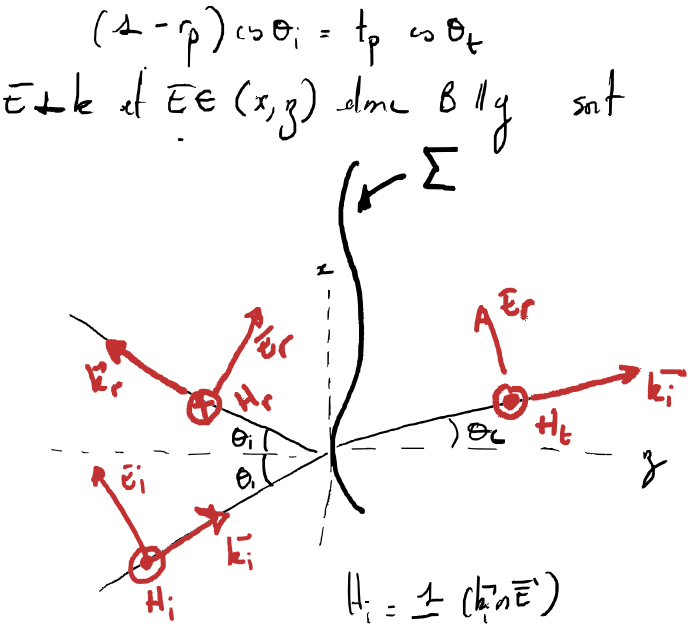
Polarisation p (transverse magnétique) : Le champ électrique est dans le plan incidence (x,z) donc les composantes tangentielles récupèrent des projection en cos θ.

(*) devient $\left(\vec{E_i}(I,t)+\vec{E_r}(I,t)\right)\cos\theta_i = \vec{E_t}(I,t)\cos\theta_t \ \forall I\in\Sigma$

or $\vec{E_r}(I,t) = r_p\vec{E_i}(I,t)$ et $\vec{E_t}(I,t) = t_p\vec{E_i}(I,t)$

soit $(1-r_p)\cos\theta_i = t_p\cos\theta_t$

et $E \perp k$ et $E \in (x,z)$ donc $B \parallel y$ soit



$H_i = \frac{1}{\mu_0\omega}(k_i\wedge E)$

$$\left\{\begin{array}{l}H_i - H_r = H_t \\ (E_i+E_r)\cos\theta_i = E_t\cos\theta_t\end{array}\right. \forall I\in\Sigma \Rightarrow \left\{\begin{array}{l}k_1(1-r_{TM}) = k_2 t_{TM} \\ (1+r_{TM})\cos\theta_i = t_{TM}\cos\theta_2\end{array}\right.$$

d'où avec $k = n\omega/c$

$$r_{TM} = \frac{n_1\cos\theta_t - n_2\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_i} \quad \text{et} \quad t_{TM} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_1}$$

$$r_{TM} = 0 \Leftarrow n_2^2\cos^2\theta_i = n_1^2\cos^2\theta_t = n_1^2(1-\sin^2\theta_t) \overset{n_1\sin\theta_i = n_2\sin\theta_t}{=} n_1^2\left(1-\frac{n_1^2}{n_2^2}\sin^2\theta_i\right)$$

$$\left(\frac{n_2^2}{n_1^2}\cos^2\theta_i = 1-\frac{n_1^2}{n_2^2}\sin^2\theta_i\right)\Big/\cos^2\theta_i \qquad \left\{\frac{1}{\cos^2\theta_i} = 1+\tan^2\theta_i\right.$$

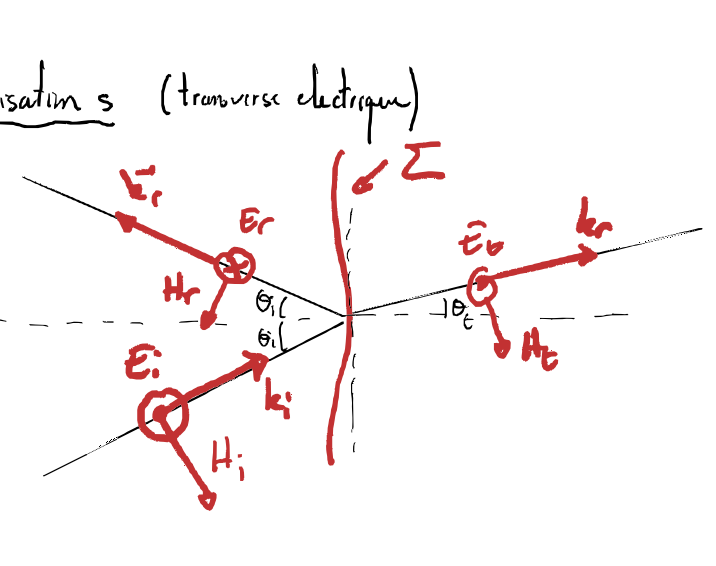
$$\frac{n_2^2}{n_1^2} = \left(1+\tan^2\theta_i\right) - \frac{n_1^2}{n_2^2}\tan^2\theta_i$$

$$1-\frac{n_2^2}{n_1^2} = \left(\frac{n_1^2}{n_2^2}-1\right)\tan^2\theta_i$$

$$\Rightarrow \tan^2\theta_i = \left(\frac{n_2}{n_1}\right)^2$$

$$\Rightarrow \theta_B = \arctan\frac{n_2}{n_1}$$ (Angle de Brewster) plus de réflexion r_{TM}

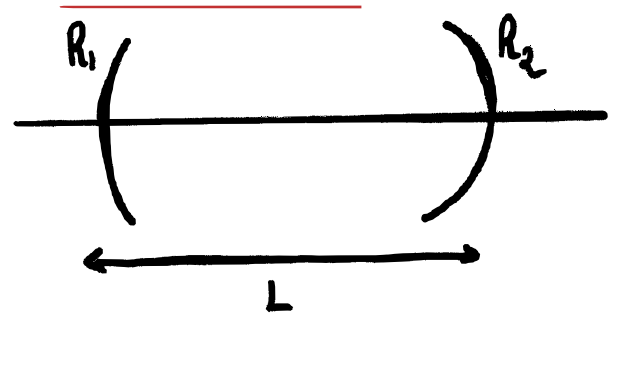
Polarisation s (transverse électrique)



$$\left\{\begin{array}{ll}E_i+E_r = E_t & I\in\Sigma \\ -H_i\cos\theta_i + H_r\cos\theta_i = -H_t\cos\theta_t\end{array}\right. \Rightarrow \left\{\begin{array}{l}1+r_{TE} = t_{TE} \\ k_{iz}(1-r_{TE}) = k_{tz}t_{TE}\end{array}\right.$$

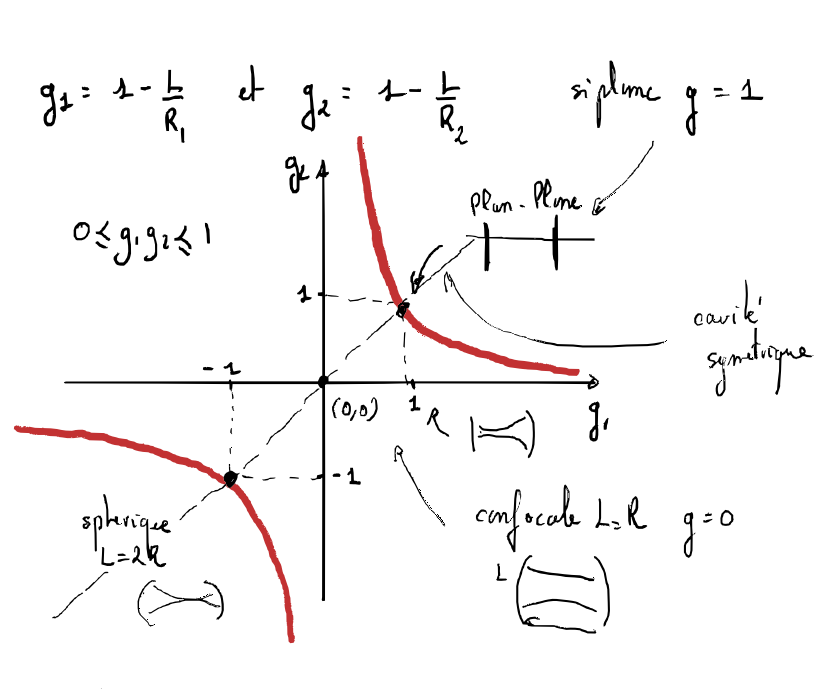
$$\left\{\begin{array}{l}r_{TE} = \frac{k_{iz}-k_{tz}}{k_{iz}+k_{tz}} \\ t_{TE} = \frac{2k_{iz}}{k_{iz}+k_{tz}}\end{array}\right. \Rightarrow r_{TE} = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t} \quad t_{TE} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t}$$

II Stabilité cavité laser



$g_1 = 1-\frac{L}{R_1}$ et $g_2 = 1-\frac{L}{R_2}$ si plane $g = 1$

$0 \le g_1 g_2 \le 1$

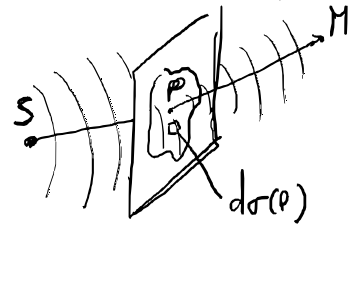


⚠ $0 \le (1-g)^2 \le 1 \Rightarrow 0 \le |1-g| \le 1$

$$\Rightarrow \left\{\begin{array}{l}0\le(1-g)\le 1 \\ \text{ou} \\ 0\le g-1\le 1\end{array}\right.$$

III Filtrage externe à la cavité

Principe de Huygens-Fresnel



$a_S(P,t) = \frac{A_0}{SP}\exp\left(j\omega t - jk_0(SP)\right)$

$da_P(M,t) = K a_S(P,t)\frac{\exp(-jk_0(PM))}{PM}\frac{t(P)d\Sigma}{d\sigma(P)}$

$= K\frac{A_0\exp(j\omega t - jk_0(SP))}{SP}\frac{\exp(-jk_0(PM))}{PM}d\sigma(P)$

$da_P(M,t) = \frac{KA_0}{SP}\frac{\exp(j\omega t)}{PM}\exp\left(-jk_0(SPM)\right)d\sigma(P)$

à grande distance $SP \simeq \infty$ et $PM \simeq \infty$ on note $C = \frac{K}{SP\,PM}$ ≈ constante (Diffraction de Fraunhofer)

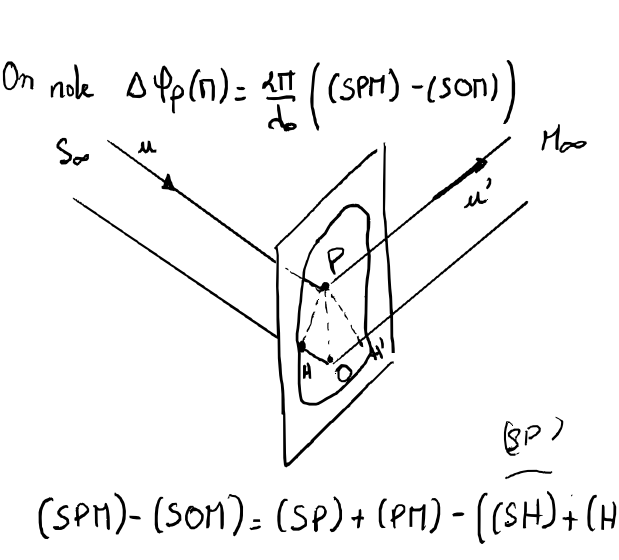
$da_P(M,t) = CA_0\exp(j\omega t)\exp\left(-jk_0(SPM)\right)\frac{t(P)d\Sigma}{d\sigma(P)}$

On note $\varphi_A(M) = \underbrace{k_0}_{2\pi/\lambda_0}(SAM)$

$da_P(M,t) = CA_0\exp(j\omega t)\exp\left(-j\varphi_P(M)\right)t(P)d\Sigma$

$= CA_0\exp(j\omega t)\exp\left(-j\varphi_0(M)\right)\exp\left(j\underbrace{\left(\varphi_P(M)-\varphi_0(M)\right)}_{\Delta\varphi_P(M)}\right)t(P)d\Sigma$

On note $\Delta\varphi_P(M) = \frac{2\pi}{\lambda_0}\left((SPM)-(SOM)\right)$



$(SPM)-(SOM) = \overset{(SP)}{\widehat{(SP)}} + \overset{(PM)}{\widehat{(PM)}} - \left((SH)+(HO)+(OH')+(H'M)\right)$

$= -(HO)-(OH') = \vec{u}\cdot\vec{OP} - \vec{u'}\cdot\vec{OP}$

$= (\vec{u}-\vec{u'})\cdot\vec{OP}$

On note $a_0(M) = A_0\exp\left(-j\varphi_0(M)\right)$

$da_P(M,t) = Ca_0(M)\exp(j\omega t)\exp\left(jk_0(\vec{u'}-\vec{u})\cdot\vec{OP}\right)t(P)d\Sigma$

on note $\vec{u} = \begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix}$ et $\vec{u'} = \begin{pmatrix}\alpha'\\ \beta'\\ \gamma'\end{pmatrix}$ et $\vec{OP} = \begin{pmatrix}x\\ y\\ 0\end{pmatrix}$

$da_P(M,t) = Ca_0(M)\exp(j\omega t)\exp\left(jk_0\left((\alpha'-\alpha)x + (\beta'-\beta)y\right)\right)t(P)d\Sigma$

$a(M,t) = \iint_\Sigma da_P(M,t) = Ca_0(M)\exp(j\omega t)\iint_\Sigma e^{jk\left((\alpha'-\alpha)x+(\beta'-\beta)y\right)}t(x,y)\,dK$

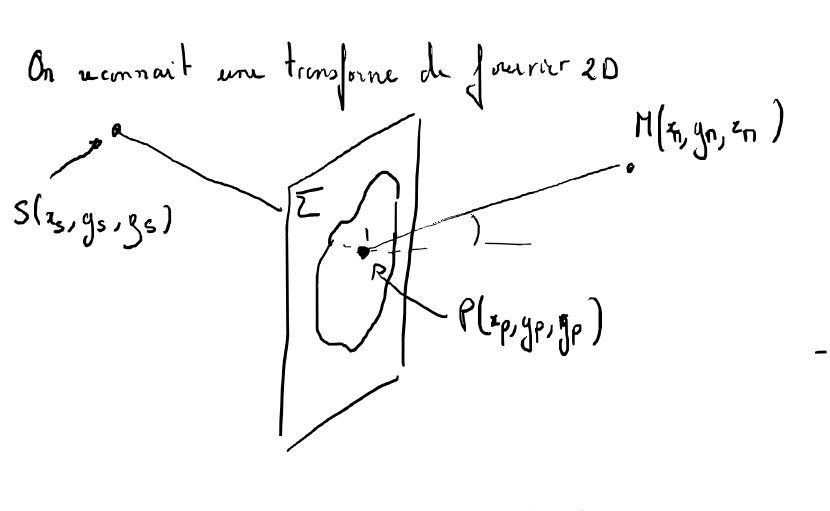
Remarque transformation de Fourier $\mathcal{F}$

$\mathcal{F}[f](k) = \alpha\int dx\, f(x)e^{\mp i\gamma kx}$, $\mathcal{F}^{-1}[g](k) = \beta\int dx\, g(x)e^{\pm i\gamma kx}$

avec $\gamma = 2\pi\alpha\beta$

Or $a(M,t) = Ca_0(M)\exp(j\omega t)\iint_\Sigma t(x,y)e^{j\left[(\alpha'-\alpha)x+(\beta'-\beta)y\right]}dx\,dy$

On reconnait une transformée de Fourier 2D



$\alpha' = \frac{x_P - x_S}{\sqrt{(x_P-x_S)^2+(y_P-y_S)^2+(z_P-z_S)^2}}$ $\beta' = \frac{y_P - y_S}{\sqrt{(x_P-x_S)^2+(y_P-y_S)^2+(z_P-z_S)^2}}$

$\gamma' = \frac{z_P - z_S}{\sqrt{(x_P-x_S)^2+(y_P-y_S)^2+(z_P-z_S)^2}}$

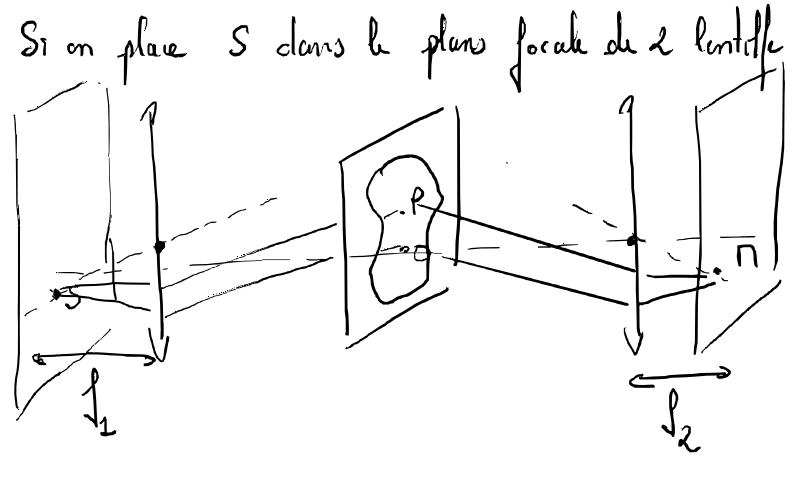
$\alpha = \frac{x_M - x_P}{\sqrt{(x_M-x_P)^2+(y_M-y_P)^2+(z_M-z_P)^2}}$ $\beta = \frac{y_M - y_P}{\sqrt{(x_M-x_P)^2+(y_M-y_P)^2+(z_M-z_P)^2}}$

$\gamma = \frac{z_M - z_P}{\sqrt{(x_M-x_P)^2+(y_M-y_P)^2+(z_M-z_P)^2}}$

S et M proche de l'axe optique et on note $P = (0,0,0)$

$\alpha' \simeq -\frac{x_S}{z_S}$ $\beta' \simeq -\frac{y_S}{z_S}$ $\gamma' \simeq 1$ $\alpha \simeq \frac{x_M}{z_M}$, $\beta \simeq \frac{y_M}{z_M}$; $\gamma \simeq 1$

Si on place S dans le plans focale de 2 lentille



alors $\alpha' = -\frac{x_S}{f_1}$; $\beta' = -\frac{y_S}{f_1}$; $\alpha = \frac{x_M}{f_2}$; $\beta = \frac{y_M}{f_2}$

Quelque exemple de transforme de fourier

- On définie la distribution de Dirac δ tel que

$$\int\delta(y)e^{\mp i\gamma yx}dy = 1$$

Donc avec la définition de la transformée de fourier $\mathcal{F}$

$\mathcal{F}[f](k) = \alpha\int f(x)e^{\mp i\gamma kx}dx$; $\mathcal{F}^{-1}[f](k) = \beta\int f(x)e^{\pm i\gamma kx}dx$

avec $\gamma = 2\pi\alpha\beta$

$\boxed{\begin{array}{ll}\mathcal{F}^{-1}[\delta]: x\mapsto\beta & \text{et } F[x\mapsto\beta] = \delta \\ F[\delta]: x\mapsto\alpha & F^{-1}[x\mapsto\alpha] = \delta\end{array}}$ donc $\boxed{\frac{1}{2\pi}\int e^{\mp iyx}dx = \delta(y)}$

- Convolution et multiplication

$(f*g)(x) = \int f(x')g(x-x')dx'$

$$\mathcal{F}[f*g]: k\mapsto\alpha\int dx\int f(x')g(x-x')dx'e^{\mp i\gamma kx} \overset{\substack{u=x-x'\\ dx=du}}{=} \alpha\int du\, g(u)e^{\mp i\gamma ku}\int f(x')e^{\mp i\gamma kx'}dx'$$

$$= \frac{1}{\alpha}F[f]\cdot F[g]$$

Donc $\boxed{\begin{array}{l}F[f*g] = \frac{1}{\alpha}F[f]\cdot F[g] \\ \text{et } F^{-1}[f*g] = \frac{1}{\beta}F^{-1}[f]\cdot F^{-1}[g]\end{array}}$

$$\mathcal{F}[f]*\mathcal{F}[g]: k\mapsto\int dy\,\alpha\int dx\, f(x)e^{\mp iyx}\,\alpha\int dx'g(x')e^{\mp i\gamma(k-y)x'}$$

$$= \alpha^2\int dx\, f(x)\int dx'g(x')e^{\mp i\gamma kx'}\underbrace{\int dy\, e^{\mp i\gamma y(x-x')}}_{\frac{2\pi}{\gamma}\delta(x-x')}$$

$$= \alpha\frac{\alpha 2\pi}{\gamma}\int dx\, f(x)g(x)e^{\mp i\gamma kx}$$

$$= \frac{1}{\beta}F[f\cdot g](k)$$

Donc $\boxed{\begin{array}{l}F[f\cdot g] = \beta F[f]*F[g] \\ F^{-1}[f\cdot g] = \alpha F^{-1}[f]*F^{-1}[g]\end{array}}$

$$\boxed{\begin{array}{l}\mathcal{F}[x\mapsto f(x+a)](x) = \left(x\mapsto e^{\pm i\gamma xa}\mathcal{F}[f](x)\right) \\ \mathcal{F}[x\mapsto f(ax)](x) = \left(x\mapsto\frac{1}{a}\mathcal{F}[f]\left(\frac{x}{a}\right)\right)\end{array}}$$

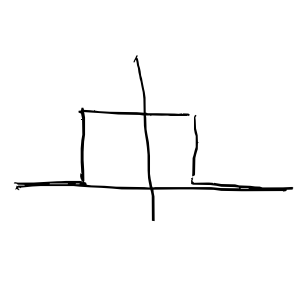
- Peigne de Dirac

$Ш_T(x) = \sum_{n\in\mathbb{Z}}\delta(x-Tn)$ $\quad\mathcal{F}[Ш_T] = \frac{1}{\beta T}Ш_{1/T}$

$\sum_{n\in\mathbb{Z}}e^{i\gamma xnT} = \frac{2\pi}{\gamma T}\sum_{m\in\mathbb{Z}}\delta\left(k-\frac{2\pi m}{\gamma T}\right)$ ← Poisson

- Porte

$$\Pi(x) = \left\{\begin{array}{ll}0 & \text{si } x^2>1 \\ \frac{1}{2} & \text{si } x^2=1 \\ 1 & \text{si } x^2<1\end{array}\right.$$



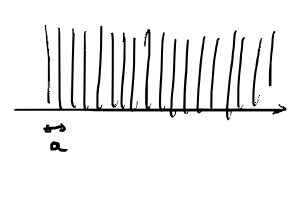
$$\mathcal{F}[\Pi](k) = \alpha\int_{-1}^{1}e^{\mp i\gamma kx}dx = \frac{\alpha}{\mp i\gamma k}\left(e^{\mp i\gamma k} - e^{\pm i\gamma k}\right)$$

$$= \alpha 2\,\mathrm{sinc}(\gamma k)$$

$\mathcal{F}[\Pi](x) = 2\alpha\,\mathrm{sinc}(\gamma x)$ avec $\mathrm{sinc}(x) = \frac{\sin(x)}{x}$

$\mathcal{F}\left[x\mapsto\Pi\left(\frac{x}{\ell}\right)\right](x) = \ell 2\alpha\sin(\gamma\ell x)$

Diaphragme composé d'une infinité de fente infiniment fines



$f_1(x) = Ш_a(x) = \frac{1}{a}Ш\left(\frac{x}{a}\right)$

$\downarrow$TF $\quad\downarrow$TF

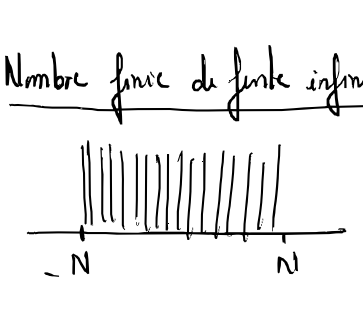
$\frac{1}{a\beta}Ш_{\frac{1}{a\beta}}(x) = \alpha Ш(a\alpha\beta x)$; $\tilde{f}_1(x)$

$\delta\left(\frac{x}{a}-n\right) = a\delta(x-an)$

$\mathcal{F}[Ш](x) = \frac{1}{\beta}\sum_m\delta\left(x-\frac{m}{\alpha\beta}\right)$ $\quad\mathcal{F}\left[x\mapsto\frac{1}{a}Ш\left(\frac{x}{a}\right)\right] = \mathcal{F}[Ш](ax) = \alpha Ш(a\alpha\beta x) = \frac{\alpha}{a\alpha\beta}Ш_{\frac{1}{a\alpha\beta}}(x)$

$= \frac{1}{\beta}\sum\alpha\beta\,\delta(\alpha\beta x - m) = \alpha Ш(\alpha\beta x)$

Nombre fini de fente infiniment fine



$f_2(x) = \frac{1}{a}Ш\left(\frac{x}{a}\right)\cdot\Pi\left(\frac{x}{(N+\frac{1}{2})a}\right)$

$\mathcal{F}\left[x\mapsto\Pi\left(\frac{x}{(N+\frac{1}{2})a}\right)\right](x) = (2N+1)a\alpha\,\mathrm{sinc}\left(\gamma(N+\tfrac{1}{2})ax\right)$

$f_2(x)\xrightarrow{TF}\beta\tilde{f}_1 * \left(x\mapsto(2N+1)\alpha\,\mathrm{sinc}\left(\gamma(N+\tfrac{1}{2})ax\right)\right)$

$\alpha Ш(a\alpha\beta x) = \alpha\sum_{n\in\mathbb{Z}}\delta(a\alpha\beta x - n) = \frac{1}{a\beta}\sum_{n\in\mathbb{Z}}\delta\left(x-\frac{n}{a\alpha\beta}\right)$

$= (2N+1)\alpha\sum_{n\in\mathbb{Z}}\mathrm{sinc}\left(\gamma(N+\tfrac{1}{2})a\left(x-\frac{n}{a\alpha\beta}\right)\right)$

$\tilde{f}_2(x) = (2N+1)\alpha\sum_{n\in\mathbb{Z}}\mathrm{sinc}\left(\gamma(N-\tfrac{1}{2})\left(ax-\frac{n}{\alpha\beta}\right)\right)$

or aussi $f_2(x) = \sum_{n=-N}^{n=N}\delta(x-an)$

donc $\tilde{f}_2(x) = \alpha\sum_{n=-N}^{N}e^{\pm i\gamma nxa}$

$$\left\{\begin{array}{l}\text{somme geometrique Cas } q\neq 1 : \sum_{n=b}^{a}q^n = q^b\frac{1-q^{a-b+1}}{1-q} = \frac{q^{a+1}-q^b}{q-1} \\ \text{Cas } q=1 \quad \sum_{n=b}^{a}1 = a-b+1\end{array}\right.$$

$$\tilde{f}_2(x) = \alpha\sum_{n=-N}^{N}\left(e^{\pm i\gamma xa}\right)^n = \alpha\frac{e^{\mp i\gamma xaN} - e^{\pm i\gamma xa(N+1)}}{1-e^{\pm i\gamma xa}}$$

$$= \alpha e^{\mp i\gamma xN}\frac{1-e^{\pm i\gamma xa(2N+1)}}{1-e^{\pm i\gamma xa}}$$

$$= \alpha\underbrace{e^{\mp i\gamma xN}\frac{e^{\pm i\gamma xa(2N+1)/2}}{e^{\pm i\gamma xa/2}}}_{1}\frac{e^{\mp i\frac{\gamma}{2}xa(2N+1)} - e^{\pm i\frac{\gamma}{2}xa(2N+1)}}{e^{\mp i\frac{\gamma}{2}xa} - e^{\pm i\frac{\gamma}{2}xa}}$$

$$= \alpha\frac{\sin\left((2N+1)\frac{\gamma}{2}ax\right)}{\sin\left(\frac{\gamma}{2}ax\right)}$$

Donc $\tilde{f}_2(x) = \alpha(2N+1)\sum_{n\in\mathbb{Z}}\mathrm{sinc}\left(\frac{\gamma}{2}(2N+1)\left(ax-\frac{n}{\alpha\beta}\right)\right)$

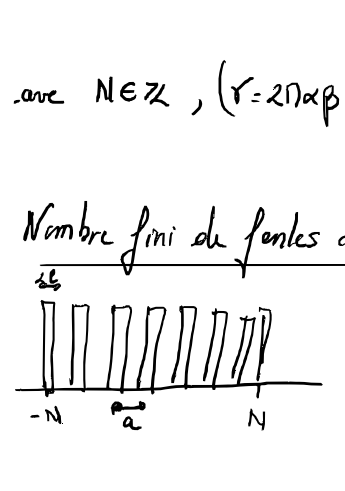
$= \alpha\frac{\sin\left(\frac{\gamma}{2}(2N+1)ax\right)}{\sin\left(\frac{\gamma}{2}ax\right)}$

Démontrons par la même occasion

$$\sum_{n\in\mathbb{Z}}\mathrm{sinc}\left(\pi(2N+1)(a\alpha\beta - n)\right) = \frac{1}{(2N+1)}\frac{\sin\left(\frac{\gamma}{2}(2N+1)ax\right)}{\sin\left(\frac{\gamma}{2}ax\right)}$$

avec $N\in\mathbb{Z}$, $(\gamma = 2\pi\alpha\beta, a)\in\mathbb{R}^2$

Nombre fini de fentes de dimension finie



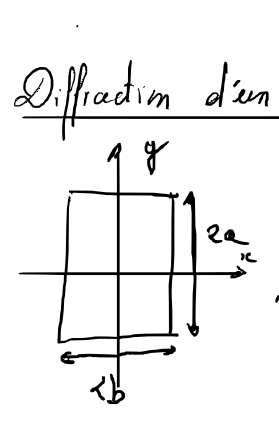
$f_3(x) = \sum_{n=-N}^{N}\Pi\left(\frac{x-na}{\ell}\right)$

$= \left(f_2 * \left(x\mapsto\Pi\left(\frac{x}{\ell}\right)\right)\right)(x)$

$\tilde{f}_3(x) = \frac{1}{\alpha}\tilde{f}_2(x)\cdot\ell 2\alpha\,\mathrm{sinc}(\gamma\ell x) = 2\ell\tilde{f}_2(x)\cdot\mathrm{sinc}(\gamma\ell x)$

$= 2\ell\alpha\frac{\sin\left(\frac{\gamma}{2}(2N+1)ax\right)}{\sin\left(\frac{\gamma}{2}ax\right)}$

Diffraction d'un rectangle



$f(x,y) = \Pi\left(\frac{x}{a}\right)\Pi\left(\frac{y}{b}\right)$

$\tilde{f}(x,y) = \alpha^2 4ab\,\mathrm{sinc}(\gamma ax)\,\mathrm{sinc}(\gamma by)$

Diffraction d'une Pupille circulaire

$\vec{x} = (x,y)$ $\quad f(\vec{x}) = \Pi\left(\frac{\|\vec{x}\|_2}{D/2}\right)$

$\tilde{f}(\theta,\psi) = \alpha^2\iint f(x,y)e^{\mp i\gamma(\theta x+\psi y)}dx\,dy$

$r = \|x\|_2 = \sqrt{x^2+y^2}$

$= \alpha^2\int_0^\infty r\,\Pi\left(\frac{r}{D/2}\right)\int_{-\pi}^{\pi}e^{\mp i\gamma\rho r\cos\varphi}d\varphi\, dr$

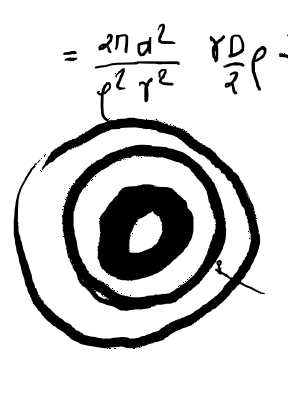
$(x,y)\mapsto(r,\varphi)$ avec $\rho = \sqrt{\theta^2+\psi^2}$ $\quad J_0(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi}e^{-ix\cos\varphi}d\varphi$

$J_1(x) = \frac{1}{i\pi}\int_{-\pi}^{\pi}e^{\mp ix\cos\varphi}\cos\varphi\, d\varphi$ et $\int_0^x yJ_0(y)dy = xJ_1(x)$

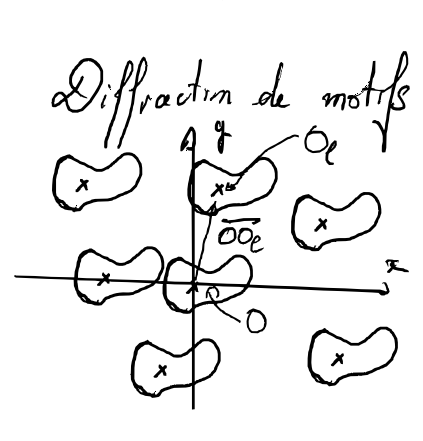
$\tilde{f}(\theta,\psi) = \frac{2\pi\alpha^2}{\rho^2\gamma^2}\int_0^{\gamma\frac{D}{2}\rho}yJ_0(y)dy$

$= \frac{2\pi\alpha^2}{\rho^2\gamma^2}\gamma\frac{D}{2}\rho J_1\left(\gamma\frac{D}{2}\rho\right) = \underbrace{\pi\alpha^2\frac{D^2}{4}}_{S}\left[\frac{2J_1\left(\gamma\frac{D}{2}\rho\right)}{\gamma\frac{D}{2}\rho}\right]$

Bessel



Diffraction de motifs identique



$f_{tot}(\vec{OP}) = \sum_{1\le\ell\le N}f(\vec{OP}-\vec{OO_\ell})$

$\tilde{E}_{top}(\vec{k}) = \alpha^2\iint f_{tot}(\vec{OP})e^{\mp i\gamma\vec{k}\cdot\vec{OP}}dS_P$

$= \alpha^2\sum_{1\le\ell\le N}\iint f(\vec{OP}-\vec{OO_\ell})e^{\mp i\gamma\vec{k}\cdot\vec{OP}}dS_P$

$\vec{OP'} = \vec{OP}-\vec{OO_\ell}$

$= \sum_{1\le\ell\le N}e^{\mp i\gamma\vec{k}\cdot\vec{OO_\ell}}\underbrace{\alpha^2\iint f(\vec{OP'})e^{\mp i\gamma\vec{k}\cdot\vec{OP'}}dS_{P'}}_{\tilde{f}(k)}$

Champ de dirac × porte forme enveloppe

$f_{tot}(\vec{OP}) = g(\vec{OP}) * f(\vec{OP})$

$\left(\vec{OP}\mapsto\sum_{1\le\ell\le N}\delta(\vec{OP}-\vec{OO_\ell})\right) * f$

or $F\left[\vec{x}\mapsto(f*g)(\vec{x})\right] = \frac{1}{\alpha^2}F[f]\cdot F[g]$ dim = 2

$F\left[\vec{x}\mapsto\delta(\vec{x})\right] = \alpha^2$

$F\left[\vec{x}\mapsto f(\vec{x}+\vec{a})\right] = \left(\vec{x}\mapsto e^{\pm i\gamma\vec{x}\cdot\vec{a}}F[f](\vec{x})\right)$

donc $F\left[\vec{x}\mapsto\delta(\vec{x}-\vec{a})\right] = \left(x\mapsto\alpha^2e^{\pm i\gamma\vec{x}\cdot\vec{a}}\right)$

donc $F\left[\vec{x}\mapsto\sum_{1\le\ell\le N}\delta(\vec{x}-\vec{OO_\ell})\right](\vec{x}) = \alpha^2\sum_{1\le\ell\le N}e^{\mp i\gamma\vec{x}\cdot\vec{OO_\ell}}$

donc $\tilde{f}_{tot} = \left(\vec{x}\mapsto\sum_{1\le\ell\le N}e^{\mp i\gamma\vec{x}\cdot\vec{OO_\ell}}\right)\cdot\tilde{f}$

Pour des motif aléatoire

Ce que l'on mesure c'est $|\tilde{f}_{tot}|^2$

et $\left|\sum_{1\le\ell\le N}e^{\mp i\gamma\vec{x}\cdot\vec{OO_\ell}}\right|^2 = \sum_{1\le\ell,m\le N}e^{\mp i\gamma\vec{x}\cdot\vec{O_mO_\ell}}$

Pour $m=\ell$ $e^{\mp i\gamma\vec{x}\cdot\vec{O_mO_\ell}} = 1$; N terme

et les reste est accompagné de son conjugué

$= N + 2\underbrace{\sum_{1\le\ell<m\le}\cos\left(\gamma\vec{x}\cdot\vec{O_mO_\ell}\right)}_{N^2-N \text{ terme de moyenne nulle}}$

$\mathcal{E}_N(M) = N\,\mathcal{E}_{motif}(N)$

IV speckles

Cohérence du laser

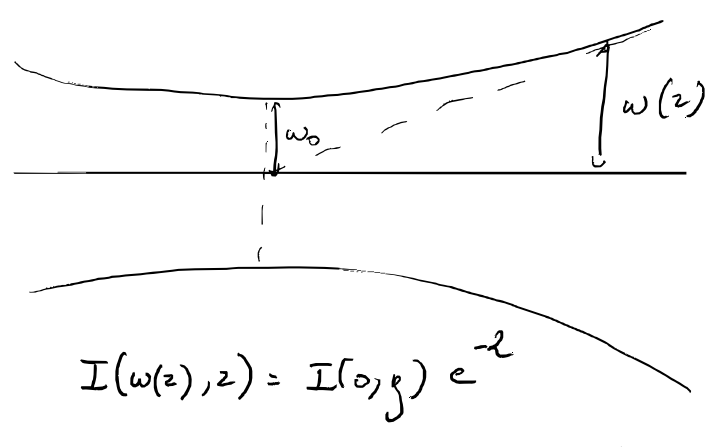
- Chaque petit point de la surface diffuse une onde
- les chemins optiques sont tous légèrement différents
- les phase deviennes aléatoire
- toutes ces ondes interfèrent

V waist

$w(z)$ ← rayon du faisceau

$I(r,z) = I_0\left(\frac{w_0}{w(z)}\right)^2e^{-2r^2/w(z)^2}$

r : distance à l'axe



$I(w(z),z) = I(0,z)e^{-2}$

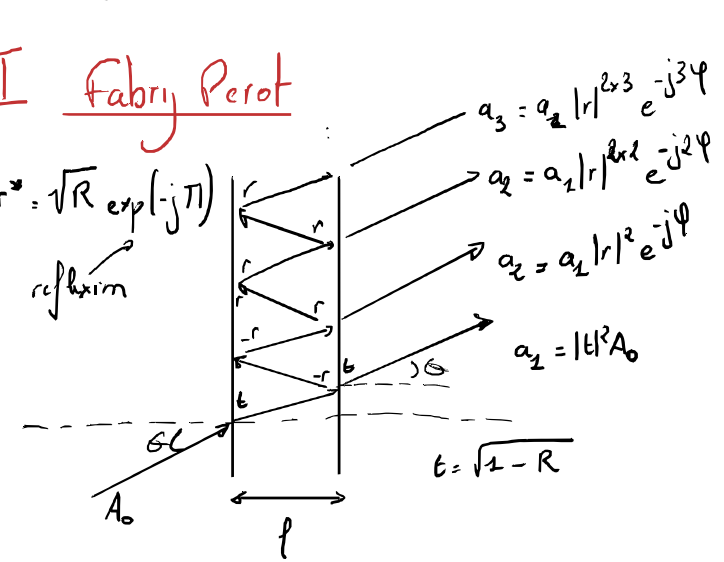
$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}$ $\quad z_R = \frac{\pi w_0^2}{\lambda}$

$w^2(z) = w_0^2 + \frac{w_0^2}{z_R^2}z^2$

affine en z^2

VI Fabry Perot

$r^* = \sqrt{R}\exp(-j\pi)$ réflexion



$a_3 = a_1|r|^{2\times3}e^{-j3\varphi}$
$a_2 = a_1|r|^{2\times2}e^{-j2\varphi}$
$a_2 = a_1|r|^2e^{-j\varphi}$
$a_1 = |t|^2A_0$
r
t
θ
A_0
ℓ
$t = \sqrt{1-R}$

La différence de marche entre a_{n+1} et a_n est $\delta = 2\ell\cos(\theta)$

donc la différence de phase est $\varphi = \frac{2\pi}{\lambda_0}2\ell\cos\theta = \frac{4\pi\ell\cos\theta}{\lambda_0}$

$a_{n+1} = a_nRe^{-j\varphi} = (Re^{-j\varphi})^na_1 = (1-R)A_0\left(R\exp(-j\varphi)\right)^n$

$a = \sum_{k=1}^{+\infty}a_k = (1-R)A_0\sum_{k=1}^{+\infty}(Re^{-j\varphi})^{k-1} = \frac{(1-R)A_0}{1-R\exp(-j\varphi)}$

$|q|<1 \rightarrow S_n = \sum_{0\le k\le n}q^k = \frac{1-q^{n+1}}{1-q}$

$a = \frac{(1-R)A_0}{1-R\cos\varphi + jR\sin\varphi} \rightsquigarrow |a|^2 = \frac{(1-R)^2\overset{E_0^2}{A_0^2}}{(1-R\cos\varphi)^2 + R^2\sin^2\varphi}$

$T = \frac{E}{E_0} = \frac{(1-R)^2}{1+R^2-2R\cos\varphi} = \frac{(1-R)^2}{(1-R)^2-4R\sin^2\frac{\varphi}{2}} = \frac{1}{1+\frac{4R}{(1-R)^2}\sin^2\frac{\varphi}{2}}$

$\cos^2\varphi + \sin^2\varphi = 1$

$\cos 2\varphi = \cos^2\varphi - \sin^2\varphi = 1-2\sin^2\varphi$

Donc $T = 1 \Leftrightarrow \sin^2\varphi/2 = 0$

$\Leftrightarrow \frac{\varphi}{2} = q\pi$

$\Leftrightarrow \frac{2\pi\ell\cos\theta}{\lambda_0} = q\pi \quad q\in\mathbb{Z}$

$\Leftrightarrow 2\ell\cos\theta = q\lambda_0 \quad q\in\mathbb{Z}$

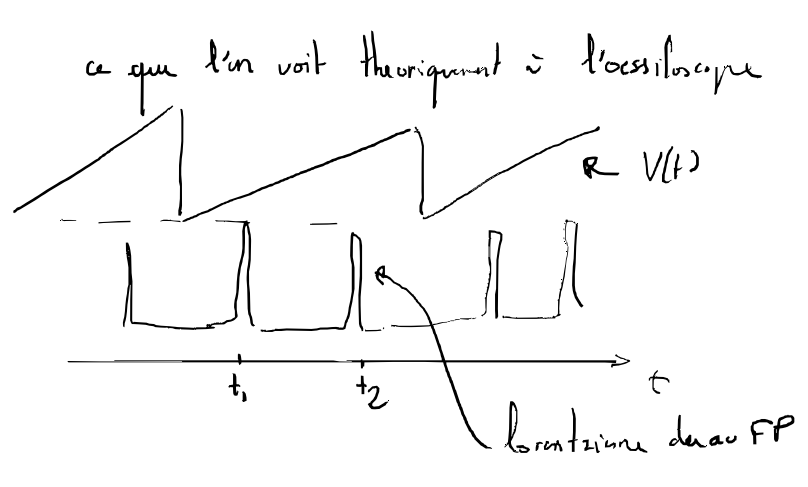
ici $\ell(t) = bt+a$ avec une fréquence $\frac{1}{T}$

Si le laser est monochromatique

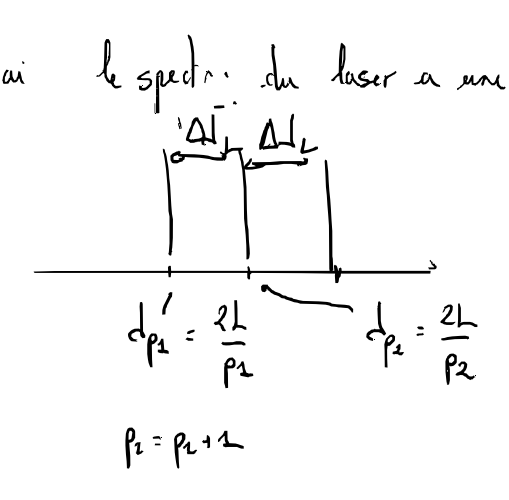
à t_1 : $2\ell(t_1)\cos\theta = q_1\lambda_0$

à t_2 : $2\ell(t_2)\cos\theta = q_2\lambda_0$

ce que l'on voit théoriquement à l'oscilloscope



Mais le spectre du laser a une structure



$p_1 = p_2+1$

Je note $\lambda_0 = \frac{(\lambda_{p_1}+\lambda_{p_2})}{2}$ et $\Delta\lambda_L = \lambda_{p_1}-\lambda_{p_2}$

$\lambda_{p_2} = \lambda_0 + \frac{\Delta\lambda_L}{2}$ et $\lambda_{p_1} = \lambda - \frac{\Delta\lambda_L}{2}$

$\nu_p = p\frac{c}{2L}$ $\quad\Delta\nu_L = \frac{c}{2L}$ et $\nu_0 = \frac{c}{\lambda_0}$

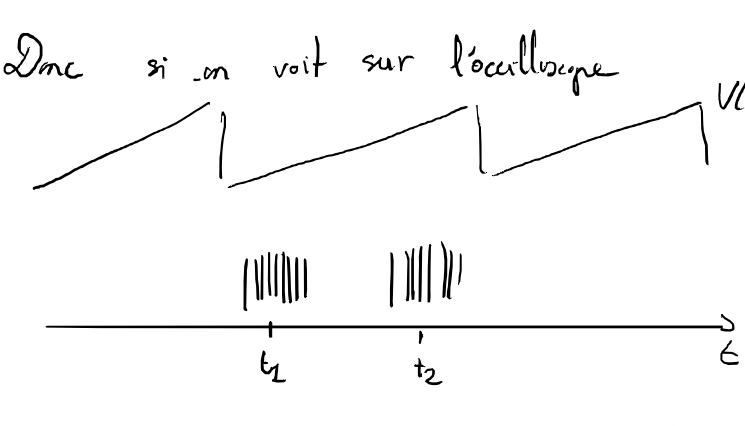
$\lambda_0 \pm \frac{\Delta\lambda_L}{2} = \frac{c}{\nu_0\mp\frac{\Delta\nu_L}{2}}$ donc

$\Delta\lambda_L = \mp 2\lambda_0 \pm\frac{2c}{\nu_0\mp\frac{\Delta\nu_L}{2}} = \mp 2c\left(\frac{1}{\nu_0} - \frac{1}{\nu_0\mp\frac{\Delta\nu_L}{2}}\right)$

$= \mp\frac{2c}{\nu_0}\left(1-\frac{1}{1\mp\frac{\Delta\nu_L}{\nu_0 2}}\right) \simeq \mp\frac{2c}{\nu_0}\left(1-1\mp\frac{\Delta\nu}{\nu_0 2}\right)$

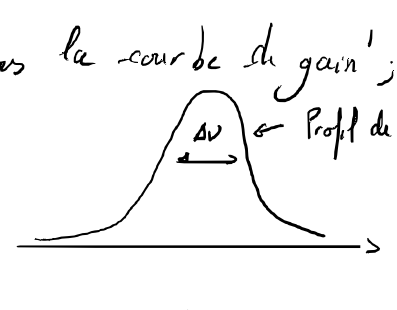
$$\boxed{\Delta\lambda_L \simeq \frac{\cancel{2}c}{\nu_0^2}\Delta\nu_L = \frac{\lambda_0}{\nu_0}\Delta\nu_L = \frac{\lambda_0}{\nu_0}\frac{c}{2L}}$$

Donc si on voit sur l'oscilloscope

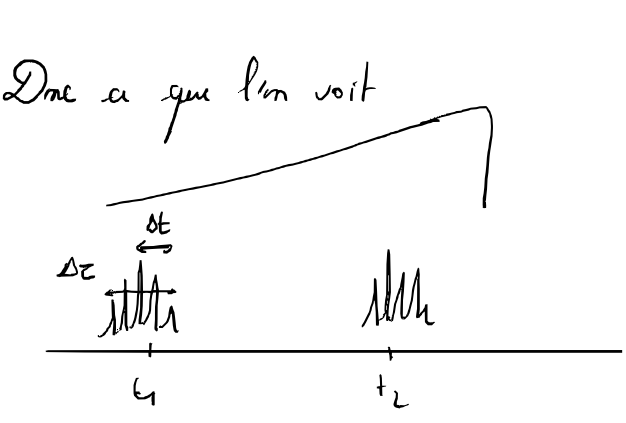


Mais pas tous les λ sont amplifier pareil

Dans la courbe de gain ;



Donc ce que l'on voit



Correspondance

ΔT → fréquence du fabry-pero 10 GHz

Δt → $\Delta\nu_L$ à trouver $\Delta\lambda_L = \frac{\lambda_0}{\nu_0}\Delta\nu_L$

Δτ → $\Delta\nu_L$

Il faut juste faire un produit en croix